Conditional Density Lift and Dual-Witness Ξ Functional

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May 2026

Version: v1.0 Pre-Print

Abstract

This paper constructs a conditional bridge between the ℓ^1 cohomological obstruction Φ_1 and a secondary trace-norm functional Ξ. The density lift is the standard ℓ^1-to-ℓ^2 square-root embedding only after fixing mass preservation, coordinatewise sign preservation, and a phase-free section; without those restrictions, the lift is not unique. The evaluation splits into two ordered diagnostics: Φ_1 **is the irreducible quotient obstruction magnitude**, while Ξ **is a deterministic tie-breaker measuring trace-norm displacement between the canonical primal residual density** $\rho(h_1^*)$ **and the canonical dual-witness density** $\rho(\phi^*)$. Ξ is not an acceptance criterion independent of Φ_1.

1 Prerequisites

All notation and results from the section-1 DSL:

$$\begin{aligned} P &:= (V, E) \\ C^0(P) &:= \mathbb{R}^V \\ C^1(P) &:= \mathbb{R}^E \\ d_0 &: C^0(P) \to C^1(P) \end{aligned}$$

From Paper 003:

$$\begin{aligned} \alpha^* &:= \operatorname{argmin}_{\alpha \in A(P)} \|\omega_1 - d_0\alpha\|_1 \\ h_1 &:= \omega_1 - d_0\alpha^* \\ \Phi_1(P, \omega_1) &:= \|h_1\|_1 \end{aligned}$$

From Paper 003, Theorem 9.1 (Cut Duality):

$$\Phi([\delta]) = \max_{\phi} \langle \phi, \delta \rangle \quad \text{subject to: } \|\phi\|_\infty \leq 1, \quad d_0^*\phi = 0$$

2 Motivation

The engine requires a secondary functional Ξ that:

1. Is derived from the ℓ^1 primal-dual solution after explicit tie-breaking
2. Measures structural quality beyond magnitude

3. Breaks ties when Φ_1 is constant
4. Does not use ℓ^2 harmonic projections as acceptance criteria
5. Does not introduce hidden degrees of freedom beyond the stated section and tie-break choices

Previous formulations used:

$$\Xi := \|\rho(h_1) - \Pi_H \rho(h_1) \Pi_H\|_1$$

This is rejected for three reasons:

1. Π_H is an ℓ^2 object reintroduced into an ℓ^1 system
2. $\Pi_H \rho \Pi_H$ is generally not a density matrix (trace may drop)
3. The construction violates the ℓ^1-primary architecture

3 Conditional ℓ^1-to-Density Lift

Given nonzero $v \in C^1(P)$, define its total ℓ^1 mass:

$$M(v) := \sum_e |v_e| = \|v\|_1$$

Define the phase-free signed amplitude vector:

$$\psi(v)_e := \operatorname{sign}(v_e) \cdot \sqrt{|v_e|/M(v)}$$

Then:

$$\sum_e |\psi(v)_e|^2 = 1$$

Define the rank-one density:

$$\rho(v) := |\psi(v)\rangle\langle\psi(v)|$$

If $v = 0$, $\rho(v)$ is undefined and any functional depending on it is defined separately by the zero-residual rule.

3.1 Properties

Trace normalization:

$$\operatorname{tr}(\rho(v)) = \sum_e |\psi(v)_e|^2 = 1$$

Positivity:

$$\forall x : \langle x, \rho(v)x \rangle = |\langle \psi(v), x \rangle|^2 \geq 0$$

Scale invariance:

$$\rho(\lambda v) = \rho(v) \quad \text{for all } \lambda \neq 0$$

Sign preservation:

$$\text{sign}(\psi(v)_e) = \text{sign}(v_e)$$

The directional information of the cochain is preserved.
Mass preservation:

$$\sum_e |\psi(v)_e|^2 = \sum_e |v_e|/M(v) = 1$$

This is the unique coordinatewise phase-free embedding satisfying the following imposed constraints:

- preserves mass (total weight = 1)
- preserves sign (directional information)
- produces a unit vector in the chosen representation space
- introduces no additional phases

Without the sign-preserving and phase-free section constraints, the lift is not unique: arbitrary coordinate phases and other representation choices produce the same descended ℓ^1 masses.

> **Status:** CONDITIONAL CONSTRUCTION. The construction is canonical only relative to the stated mass, sign, and section constraints.

4 Tie-Broken Primal Residual and Optimal Dual Witness

4.1 Existence

By Theorem 9.1 (Paper 003) and LP strong duality on finite graphs, there exists ϕ^* such that:

$$\begin{aligned} \phi^* &\in \ker(d_0^T) \\ \|\phi^*\|_\infty &\leq 1 \\ \langle \phi^*, h_1 \rangle &= \Phi_1 \end{aligned}$$

The dual constraint $d_0^T \phi = 0$ means ϕ is a divergence-free circulation in the algebraic transpose sense. The primal removes gauge; the dual maximizes over gauge-invariant probes.

4.2 Tie-Broken Primal Selection

If multiple primal minimizers exist, define:

$$R_{\text{opt}} := \{\omega_1 - d_0\alpha : \alpha \in A(P), \|\omega_1 - d_0\alpha\|_1 = \Phi_1\}$$
$$h_1^* := \text{argmin}_{r \in R_{\text{opt}}} \|r\|_2$$

If the ℓ^2 tie-break is still non-unique, fix a deterministic lexicographic rule relative to a declared edge ordering. This tie-break is representational; it does not alter the primary obstruction value Φ_1.

4.3 Tie-Broken Dual Selection

If multiple optimal ϕ^* exist, select canonically:

$$\phi^* := \text{argmin}_{\phi \in \text{Opt}} \|\phi\|_2$$

where $\text{Opt} = \{\phi : d_0^T\phi = 0, \|\phi\|_\infty \leq 1, \langle \phi, h_1^* \rangle = \Phi_1\}$.

This ℓ^2 use is strictly a deterministic tie-breaking rule for representation. It is not an acceptance functional and does not enter the descent law.

4.4 Interpretation

The dual witness ϕ^* is:

> an optimal gauge-invariant probe that detects the selected residual h_1^*

After tie-breaking, it identifies one circulation structure that witnesses the maximal obstruction.

> **Status:** DERIVED for existence of primal and dual optima under Paper 003's finite-dimensional assumptions. The ℓ^2 and lexicographic tie-breaks are additional deterministic representation choices.

5 The Primal-Dual Representative Distortion Ξ Functional

5.1 Definition

$$\Xi(P, \omega) := \begin{cases} 0, & \text{if } \Phi_1(P, \omega) = 0 \\ \|\rho(h_1^*) - \rho(\phi^*)\|_{\text{tr}}, & \text{otherwise} \end{cases}$$

where:

- ω is the explicit operational gauge representative field (current defect instantiation).
- h_1^* is the tie-broken primal residual selected in §4.2.
- ϕ^* is the tie-broken optimal dual witness selected in §4.3.
- $\rho(\cdot)$ is the conditional ℓ^1-to-density lift (§3).
- $\|\cdot\|_{\text{tr}}$ denotes the trace norm (Schatten-1 norm).

5.2 Properties

Non-negativity:

$$\Xi \geq 0$$

(trace norm is a norm)

Zero condition:

$$\Xi = 0 \iff \rho(h_1^*) = \rho(\phi^*)$$

For the real phase-free section, equality of rank-one densities means the normalized signed amplitude vectors agree up to global sign. This is a representative-alignment condition, not a statement that the raw field ω is harmonic.

Bounded:

$$\Xi \leq 2$$

(the Schatten-1 distance between density matrices is at most 2; the half-normalized trace distance is at most 1)

Scale invariance (inherited):

Ξ is invariant under nonzero rescaling of h_1^* and ϕ^*

(because ρ is scale-invariant)

5.3 What Ξ Measures

Ξ = geometric displacement between the canonical primal residual h_1^* and the canonical dual witness ϕ^*

Interpretation:

- Φ_1 measures **irreducible obstruction magnitude** for the quotient class induced by the chosen ℓ^1 cochain geometry. It ignores exact variation completely.
- Ξ measures **primal-dual representative distortion** after deterministic tie-breaking. It is a secondary diagnostic for alignment between the residual and its saturating circulation certificate.

When Ξ is large, the selected primal residual and selected dual certificate have substantially different normalized support/sign profiles. This does not by itself prove a larger obstruction.

5.4 Advantages Over Previous Formulation

Status: CONDITIONAL DERIVED. The functional uses only objects already present in the section-1 DSL plus Paper 003 duality, but it depends on declared deterministic tie-breaks and the phase-free density section.

Property	**Old Ξ (Π_H-based)**	**New Ξ (representative distortion)**
Required raw evaluation parameterization	Yes	No
Free phases	Yes (leak)	No
Trace-preserving	Not guaranteed	Yes (both inputs are strictly normalized density matrices)
Canonically defined	No	Conditional on explicit primal/dual tie-breaks
Evaluates valid diagnostic bounds linearly	No	Yes, as a secondary primal-dual diagnostic
External quantum score	Possible leak	Avoided by construction, provided Ξ remains secondary

6 Acceptance Law (Guarded Hierarchy)

The evaluator hierarchy used by the surrounding engine is:

$$\text{accept iff: } R_{\text{cl, after}} \leq R_{\text{cl, max}} \text{ AND } \left(\begin{array}{c} \Phi_{1,\text{after}} < \Phi_{1,\text{before}} \\ \text{OR } (\Phi_{1,\text{after}} = \Phi_{1,\text{before}} \text{ AND } \Gamma_{2,\text{after}} < \Gamma_{2,\text{before}}) \\ \text{OR } (\Phi_1, \Gamma_2 \text{ tie AND } \Xi_{\text{after}} < \Xi_{\text{before}}) \end{array} \right)$$

Ξ remains strictly secondary. It cannot authorize a step that increases Φ_1 or Γ_2, and it is ignored if the R_{cl} gate fails.

7 Structural Correspondence to Trace-Distance Duality

7.1 The Norm-Duality Pattern

Both the engine and quantum information theory share:

- **primal**: minimize over equivalence (gauge/unobservable transforms)
- **dual**: maximize over admissible probes (circulations/measurements)

Exact mapping:

Section-1 DSL	**Quantum Info**
δ	$(\rho - \sigma)$
ϕ (divergence-free)	measurement operator M
$d_0^T \phi = 0$	measurement admissibility
$\|\phi\|_\infty \leq 1$	operator norm bound $\|M\| \leq I$
$\Phi([\delta])$	formally analogous to an unhalved trace-norm distance
$\text{im}(d_0)$	gauge orbit
$\ker(d_0^T)$	observable space

7.2 Precise Status

This correspondence is:

STRUCTURAL CORRESPONDENCE (same mathematical pattern)

It is **not** an identity because:

graph circulations $\neq$ Hermitian operators

The correspondence can be promoted toward an operator-level identity only after the operator-valued lift (Paper 004):

$C^1(P) \to C^1(P, F)$ where $F(x) =$ trace-class operators

is constructed and verified.

Status: CONDITIONAL DERIVED. The structural pattern is proven. The operator-level identity requires the lift from Paper 004 §operator-valued presheaves.

8 The Core Identity

The clean statement expressible within the theory:

- **gauge freedom** $= \mathrm{im}(d_0)$
- $\Phi_1 =$ magnitude of topological obstruction class (invariant under exact modifications)
- $\Xi =$ trace-norm distortion between selected primal and dual representatives

This is Banach duality restricted to the quotient $C^1/\mathrm{im}(d_0)$:

$$\|x\|_1 = \sup_{\|\phi\|_\infty \leq 1} \phi(x) \quad \text{restricted to the algebraic annihilator } \ker(d_0^T)$$

9 What Is Proven vs What Is Open

9.1 DERIVED

1. Φ is the standard quotient norm induced by the fixed ℓ^1 norm on C^1 [Paper 003]
2. Φ has LP dual = bounded divergence-free circulations [Theorem 9.1]
3. ℓ^1 aggregation follows under the observer assumptions of Papers 001-002
4. ℓ^2 Hodge component bookkeeping cost can inflate relative to the sparse representative in the specific cycle-graph construction of Paper 003
5. The density lift $\rho(\cdot)$ is canonical only under mass preservation, sign preservation, and a phase-free section
6. The dual-witness Ξ uses only ℓ^1 primal-dual data plus stated tie-breaks

9.2 CONDITIONAL DERIVED

7. If $F : C^1(P) \to$ trace-class operators is constructed such that $\|h_1\|_1 = \|F(h_1)\|_{\text{trace}}$, then Φ corresponds to trace distance

8. Bounded circulations correspond to measurement operators only after the operator-valued lift is constructed

9.3 OPEN

9. Explicit construction of the lifting functor F. Any valid construction must satisfy:
   - Norm preservation: $\|h_1\|_1 = \|F(h_1)\|_{\text{trace}}$
   - d_0 compatibility: $F(d_0\alpha) = d_0'(F(\alpha))$
   - Dual correspondence: $\ker(d_0^*) \leftrightarrow$ measurement operators

10. Formal proof that bounded circulations $\cong$ POVMs under the lift

11. Derivation of Born rule from Φ (not claimed, not attempted)

10 Concrete Verification Plan

10.1 Test 1 — Path Graph Strong Duality

- $P = 1 \to 2 \to 3$
- $\omega_1 = [1, -1]$
- compute α^*, h_1, Φ_1
- compute ϕ^* via dual LP
- verify $\langle \phi^*, h_1 \rangle = \Phi_1$

10.2 Test 2 — Ξ Computation

- compute $\rho(h_1^*), \rho(\phi^*)$
- compute $\Xi = \|\rho(h_1^*) - \rho(\phi^*)\|_{\text{tr}}$
- verify $\Xi \geq 0$
- verify $\Xi = 0$ when $h_1^* \propto \phi^*$

10.3 Test 3 — 3-Node Cycle

- $P =$ directed 3-cycle
- $\omega_1 = [1, 1, 1]$ (uniform circulation)
- verify h_1 is the full signal (nothing is exact)
- compute dual witness ϕ^*
- verify strong duality
- compute Ξ

10.4 Test 4 — Zero Residual

- P = path graph
- $\omega_1 = d_0\alpha$ for some α
- verify $h_1 = 0$
- verify $\Phi_1 = 0$
- verify $\Xi = 0$

10.5 Test 5 — Operator Lift (when available)

- define $F : C^1(P) \to$ trace-class operators
- compute Φ via Schatten-1 norm
- compare to scalar Φ_1
- verify correspondence

A Relationship to Previous Ξ Definitions

The previous Ξ definition used:

$$\begin{aligned} \rho_H(h_1) &:= \Pi_H \rho(h_1) \Pi_H \\ \Xi_{\text{old}} &:= \|\rho(h_1) - \rho_H(h_1)\|_1 \end{aligned}$$

where Π_H is the orthogonal projection onto $\mathrm{Harm}^1(P) = \ker(d_1) \cap \mathrm{im}(d_0)^\perp$.
This is replaced because:

1. Π_H reintroduces ℓ^2 structure that Paper 003 explicitly demoted
2. $\Pi_H \rho \Pi_H$ is not necessarily trace-1 (violates density matrix requirement)
3. The harmonic subspace depends on d_1, which may not be defined at degree 1

The new definition:

$$\Xi_{\text{new}} := \|\rho(h_1^*) - \rho(\phi^*)\|_{\text{tr}}$$

uses only objects from the ℓ^1 primal-dual pair and requires no external structure.

Migration note: Existing engine implementations using Ξ_{old} should be updated. The acceptance law is unchanged; only the Ξ computation changes.

B Phase Derivation from Dual Witness (Optional Extension)

If phase information is desired (e.g., for operator-valued lifts), one deterministic derivation is:

$$\theta_e := \mathrm{sign}(\phi^*(e))$$

This assigns the phase of each edge amplitude from the dual witness direction, producing:

$$\psi_e = \mathrm{sign}(\phi^*(e)) \cdot \sqrt{|h_1^*(e)|/\Phi_1}$$

This is optional. The base construction (§3.4) uses $\theta_e = 0$ or $\mathrm{sign}(h_1^*(e))$.

The dual-witness phase derivation becomes meaningful only after the operator-valued lift of Paper 004 is constructed.

Status: OPTIONAL EXTENSION. Not required for the base Ξ functional.

Changelog

v002 (April 2026)

- Explicit integration of Conditional Minimality to formally quarantine the L2 embedding mechanism as an analytic mapping layer, preventing confusion with structural foundations.
- Replaced the inconsistent harmonic-representative Ξ definition with the primal-residual/dual-witness definition used by the appendix and verification plan.
- Downgraded density-lift uniqueness to a conditional statement requiring mass preservation, sign preservation, and a phase-free section.

v001 (April 2026)

- §3 rewritten: replaced dual-definition density lift (raw vector $\rho(v) = |v\rangle\langle v|/\langle v, v\rangle$ AND sqrt embedding) with single canonical $\ell^1 \to \ell^2$ sqrt embedding. $\psi(v)_e = \mathrm{sign}(v_e) \cdot \sqrt{|v_e|/M(v)}$ is now the only density construction.
- §5.3 tightened: "alignment" $\to$ "geometric discrepancy between primal minimizer and dual maximizer"

v000 (April 2026)

- Initial draft establishing:
  - Canonical rank-one density lift (§3) with phase resolution
  - Optimal dual witness construction from Theorem 9.1 (§4)
  - Dual-witness Ξ functional replacing Π_H-based definition (§5)
  - Structural correspondence to trace-distance duality (§7)
  - Proven/conditional/open classification (§9)
  - Concrete verification plan (§10)